

Project Subject List

Project Subject List

Reasoning Under Uncertainty · 605.745

How to Use This List

The subjects below are grouped by **reasoning framework** rather than by domain, because the framework, not the application area, is what this course is evaluating. Each subject already implies an obvious comparison point or baseline, which is a required element of every project (see the [Project Information \(https://jhu.instructure.com/courses/134899/pages/project-information\)](https://jhu.instructure.com/courses/134899/pages/project-information) page for the full requirements).

You are not limited to the specific applications listed under each framework - treat them as examples of the kind of problem that framework fits well. If you have a domain from your own work you'd like to apply one of these frameworks to, or want to propose a topic not listed here, contact me before submitting your proposal.

Probabilistic Graphical Models (Bayesian Networks)

- **Medical diagnosis from symptoms and test results** — compare against a naive Bayes classifier to show what the network's dependency structure buys you.
- **Equipment or fault diagnosis** in a mechanical or electrical system — compare against a simple rule-based fault tree.
- **Insurance or credit risk scoring** under incomplete applicant data — compare against logistic regression.
- **Spam or fraud detection** with conditionally dependent evidence — compare against a naive independence assumption to quantify the cost of ignoring correlation.

Dempster-Shafer Evidence Theory

- **Sensor fusion for object classification** (e.g., combining camera and radar evidence) — compare belief/plausibility bounds against a Bayesian posterior on the same evidence.
- **Multi-source threat or intrusion detection** with conflicting alerts — compare against simple evidence-averaging.
- **Forensic or intelligence analysis** combining unreliable witness or source reports — compare against majority voting.

Fuzzy Logic / Possibility Theory

- **Control system under imprecise sensor readings** (e.g., HVAC, cruise control, water-level control) — compare against a crisp threshold-based controller.
- **Risk assessment with linguistically expressed inputs** ("high," "moderate") — compare against a hard-coded numeric scoring rule.
- **Consumer preference or recommendation modeling** with vague criteria — compare against a hard-cutoff filtering approach.

Markov Models / Hidden Markov Models

- **Activity or gesture recognition** from sensor sequences — compare against a memoryless (single-timestep) classifier.
- **Text or DNA sequence pattern/anomaly detection** — compare against an n-gram frequency baseline.
- **Equipment degradation / predictive maintenance** modeled as hidden health states — compare against a threshold-on-raw-signal baseline.

Sequential Decision-Making (MDP / POMDP)

- **Autonomous navigation or path planning** under partial observability — compare against a greedy/reactive policy.
- **Inventory management or resource allocation** under demand uncertainty — compare against a fixed reorder-point heuristic.
- **Adaptive treatment or intervention policy** in a simulated patient model — compare against a static protocol.

State Estimation (Kalman / Particle Filters)

- **Object tracking from noisy position/velocity measurements** — compare a Kalman filter against raw sensor smoothing (e.g., moving average).
- **Robot localization** with noisy odometry and landmark observations — compare a particle filter against dead-reckoning alone.
- **Financial time-series state estimation** (e.g., latent volatility) — compare against a simple exponential smoothing model.

Open / Hybrid

- **Head-to-head comparison of two frameworks** on the same problem (e.g., a Bayesian network vs. a fuzzy inference system on the same diagnosis task) — the comparison itself is the project.
- **Propose your own topic** in your professional domain, subject to instructor approval, provided it satisfies the required elements on the Research Project page: a named framework, a working artifact, a quantitative metric, and a comparison or baseline.

Next Step

Once you've picked a subject, move on to the Project Proposal deliverable described on the [Project Information](https://jhu.instructure.com/courses/134899/pages/project-information) (<https://jhu.instructure.com/courses/134899/pages/project-information>)_page.